

Simulation Methods in Medical Engineering
CM2014

Computer Laboratory Assignment G1

- Ultrasound image simulation -

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A) Laboratory Assignments

Your task in this lab is to perform ultrasound image simulations of a simple cyst phantom using the Matlab-based Field II ultrasound simulator. The phantom you will be working with contains ten point targets separated by 5 mm and a 10 mm water-filled cyst. All scatterers are situated in a box of (x,y,z)=(40,10,50) mm.

1) Download and unzip the "FieldII_CM2014_G1" folder from Canvas (F: Medical imaging simulations).

2) Run `field_init.m` from the downloaded folder to initiate the Field II simulator.

3) Generate the cyst phantom by executing the function `[positions, amp] = cyst_phantom(N)`. Analyze the code and output variables to understand how the phantom is generated. For example, 5000 point scatterers can be used.

4) Simulate an ultrasound image using the script `sim_cyst_phantom`. Analyze the code and the simulated image. Identify some of the imaging setup configurations. Note that the number of scatterers in the phantom is now set in `sim_cyst_phantom` when calling `[positions, amp] = cyst_phantom(N)`.

5) Elaborate on different scatter densities. How is the simulated image affected?

6) Determine the imaging focus depth (line 18 and 44 in `sim_cyst_phantom`). Does it match the focus interpreted when viewing the simulated image?

7) Change the focus depth and verify it in simulations.

8) Modify the cyst phantom in any way you like (e.g.: add a specular reflection, add a second cyst). Verify your changes to the phantom in the simulations.

B) Laboratory report Instructions (F1-Ultrasound)

The lab report should be written individually and answer the following questions. Note that you do not have to account for all the tasks you performed during the lab.

1) Briefly explain the basic theory of ultrasound image simulation
2) For what purpose can ultrasound image simulations be useful? Please give some examples.
3) Describe the dependence of point scatterer density in the simulations. Illustrate with example images from the lab. How many scatterers are needed in the phantom to simulate fully developed speckles?
4) Describe the difference between a specular reflection and backscattering. How do they show up in the image?
5) Describe how changing the focus depth affected the simulations. Illustrate using example images from the lab.
6) Account for your modifications in Lab assignment 8. Were you able to verify them in simulations?